DESIGN AND IMPLEMENTATION OF ECG SIGNAL ACQUISITION, FILTERING AND DIGITAL PULSE COUNTING SYSTEM

A Project Report Submitted in Partial Fulfillment of the Requirements

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IN

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

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COEP Technological University
Academic Year: 2025–26

CERTIFICATE

This is to certify that the project report entitled

"Design and Implementation of ECG Signal Acquisition, Filtering and Digital Pulse Counting System"

is a Bonafide work carried out by

Pulkit Dhingra

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in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering at College of Engineering Pune (COEP).

This work has been carried out under my supervision and guidance.

Date: ____________

Guide Signature: ____________
Dr. Deeplaxmi V. Niture

Head of Department: ____________

DECLARATION

We hereby declare that the project entitled

"Design and Implementation of ECG Signal Acquisition, Filtering and Digital Pulse Counting System"

is our original work and has been carried out by us.

This work has not been submitted previously for any degree or diploma in any institute or university.

We further declare that all sources of information used in this project have been properly acknowledged.

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ABSTRACT

The electrocardiogram (ECG) is one of the most widely used non-invasive diagnostic tools for assessing cardiac function. However, the raw ECG signal acquired from surface electrodes is very small in amplitude (typically 0.5–5 mV) and is heavily corrupted by power-line interference, electrode contact noise, motion artifacts and high-frequency electromagnetic interference. As a result, careful analog signal conditioning is essential before the signal can be used for heart-rate computation or display.

This project presents the design, simulation and hardware implementation of a complete ECG signal conditioning chain together with a digital pulse counting and display subsystem. The analog front-end employs an AD620 instrumentation amplifier with a calculated gain of $G \approx 97.86$ ($R_g = 510\ \Omega$), a first-order active high-pass filter with a cutoff frequency $f_L \approx 0.048$ Hz to block the DC electrode offset, and a first-order active low-pass filter with $f_H \approx 50$ Hz ($R = 318\ k\Omega$, $C = 10$ nF) to attenuate high-frequency EMI and muscle noise. Together these stages form a band-pass response that preserves the diagnostic ECG bandwidth.

The digital subsystem converts the conditioned ECG signal into a digital pulse train using an LM393 voltage comparator ($V_{ref} \approx 1.00$ V), and counts the R-peaks over a one-minute window. An NE555 timer in monostable mode ($R = 545\ k\Omega$, $C = 100\ \mu F$, $T \approx 60$ s) defines the counting window, and two cascaded CD4026 decade counter / 7-segment driver ICs display the two-digit beats-per-minute (BPM) value.

The complete system was simulated in Proteus and implemented on a breadboard. Real laboratory CRO measurements on the Rigol MSO5204 confirm correct operation at every stage: the instrumentation amplifier output is centred at ≈60 mV with $V_{pp} \approx 1.90$ V; the HPF output preserves the signal ($V_{pp} \approx 1.86$ V at 10 Hz);

the LPF output shows the expected mild attenuation ($V_{pp} \approx 1.80$ V); and the comparator generates a clean 5.04 V TTL-compatible pulse train at the same input frequency. The dual 7-segment display shows the cumulative count over the 60-second window, demonstrating end-to-end functionality.

Keywords: ECG, AD620 instrumentation amplifier, active filter, NE555 monostable, CD4026 decade counter, 7-segment display, biomedical signal conditioning.

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ABBREVIATIONS AND NOMENCLATURE

Symbol / Abbreviation	Meaning
ECG	Electrocardiogram
IA	Instrumentation Amplifier
HPF	High-Pass Filter
LPF	Low-Pass Filter
BPF	Band-Pass Filter
CMRR	Common-Mode Rejection Ratio
EMI	Electromagnetic Interference
RFI	Radio-Frequency Interference
TTL	Transistor–Transistor Logic
BPM	Beats Per Minute
LSD / MSD	Least / Most Significant Digit
MR	Master Reset (CD4026)
INH	Clock Inhibit (CD4026)
DEI / DEO	Display Enable In / Out (CD4026)
CO	Carry Out (CD4026)
G	Voltage gain (dimensionless)
f_c, f_L, f_H	Cutoff / lower / upper cutoff frequency (Hz)
R_g	Gain-setting resistor of AD620 (Ω)
V_ref	Comparator reference voltage (V)
V_pp	Peak-to-peak voltage (V)

Chapter 1 Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide and is responsible for nearly one in three deaths globally. A significant fraction of these deaths arise from arrhythmias and other cardiac events that strike with little prior warning. Early detection, continuous monitoring and even simple heart-rate measurement therefore play a central role in clinical diagnosis as well as in low-cost home health screening.

The electrocardiogram (ECG) is the most accessible and widely used non-invasive method of observing cardiac electrical activity. By placing surface electrodes on the skin and measuring the small potential differences generated by the heart's electrical cycle, an ECG provides direct information about heart rate, rhythm and the integrity of the cardiac conduction system. The ECG waveform is conventionally described by the P, Q, R, S and T deflections, with the R-peak being the largest and most easily detectable feature.

From an instrumentation viewpoint, however, the ECG is an exceptionally challenging signal to acquire. Its amplitude is on the order of 0.5–5 mV at the electrodes, its useful bandwidth is roughly 0.05–100 Hz, and the signal is invariably contaminated by 50 Hz power-line interference, half-cell electrode DC offset, motion and respiration artefacts, muscle (EMG) noise and radio-frequency pick-up. A well-engineered analog front-end is therefore the cornerstone of any meaningful ECG application.

1.2 Need for ECG Signal Conditioning

A typical raw ECG signal at the electrode–amplifier interface presents three simultaneous problems:

- It is too small to be processed by digital electronics directly — a gain of approximately 60–120 V/V is required to bring it into the comfortable working range of a 5 V system.
- It contains a slowly drifting DC offset (±300 mV) caused by the silver/silver-chloride electrode–electrolyte half-cell potential, which would otherwise saturate the amplifier.
- It carries high-frequency noise from muscle activity, radio interference and switching electronics, much of which lies above 50 Hz and outside the diagnostic band.

These three issues call for a high-CMRR differential amplifier, a high-pass filter with a very low cutoff to remove the DC offset, and a low-pass filter to attenuate high-frequency noise —

in other words, a band-pass conditioning chain. The instrumentation amplifier (IA) is preferred over an ordinary op-amp because it provides both differential gain and excellent rejection of common-mode interference (most notably 50 Hz mains pick-up coupled equally into both electrodes).

1.3 Need for Pulse Counting and Display

Beyond simply observing the ECG waveform on a scope, the most useful single quantity extracted from it is the heart rate, expressed in beats per minute (BPM). Counting R-peaks over a fixed time window of 60 seconds yields the BPM directly, with no division or scaling required. The challenge is therefore reduced to two practical sub-problems: detecting each R-peak as a clean digital pulse, and counting these pulses for exactly one minute on a numerical display.

The digital subsystem in this project performs exactly this task. A voltage comparator converts each R-peak into a TTL-compatible pulse, an NE555 in monostable mode opens a one-minute counting window, and two cascaded CD4026 decade counters drive a pair of 7-segment displays to show the running count from 00 to 99 — a range that comfortably covers the normal human resting heart rate (60–100 BPM) and most tachycardic conditions of clinical interest.

1.4 Project Objectives

The principal objectives of this work are:

1. To design and analytically size an analog front-end consisting of an instrumentation amplifier, an active high-pass filter and an active low-pass filter, that together implement a band-pass response covering the diagnostic ECG bandwidth.
2. To design a digital pulse-counting subsystem capable of generating a one-minute counting window, detecting heart pulses and displaying the count on a two-digit 7-segment display.
3. To simulate the analog and digital sub-circuits in Proteus and verify their stage-by-stage behaviour.
4. To implement the complete system on a breadboard using readily available components and to validate its operation by stage-wise CRO measurements.
5. To analyse the practical sources of error and to comment on improvements that would be necessary for clinical-grade operation.

1.5 Scope of the Project

The scope of the present work is deliberately limited to a single-lead ECG acquisition system with two-digit BPM display. The system is intended as an educational demonstrator and is not certified for clinical use. The patient-isolation, defibrillator-protection and right-leg-driven shield circuitry typical of medical-grade ECG instruments are outside the scope of this report; however, the architecture chosen is fully compatible with such extensions.

1.6 Organization of the Report

Chapter 2 presents the complete system block diagram together with the functional role of each block. Chapter 3 reviews the reference literature and explains the modifications introduced in our design. Chapter 4 covers the detailed analog design and Chapter 5 the digital design. Chapter 6 describes the hardware implementation and the practical issues encountered. Chapter 7 presents and discusses the measured results, and Chapter 8 concludes the report with a summary of achievements and a clear statement of future scope. Detailed calculations, additional CRO captures and a complete bill of materials are placed in the appendices.

Chapter 2 System Overview and Block Diagram

2.1 Overall System Architecture

The complete instrument can be partitioned into two cleanly separated sections that communicate through a single, well-defined node — the filtered ECG output. The analog section is responsible for acquiring the raw biopotential signal, amplifying it with high common-mode rejection and shaping its frequency content to the diagnostic ECG bandwidth. The digital section is responsible for converting the resulting waveform into a clean pulse train, counting the pulses for a fixed time window and displaying the count.

This two-section partition is deliberate. It keeps the noisy high-impedance analog circuitry physically and electrically separate from the fast-switching digital circuitry, which is essential to prevent digital ground bounce and switching transients from coupling back into the millivolt-level ECG signal. The analog and digital sections in our implementation share a common ground and a common +5 V supply, but they were built on physically separate breadboards for exactly this reason.

2.2 Complete System Block Diagram

Figure 2.1 shows the end-to-end signal path of the system. Each block is labelled with its principal function. The arrows indicate the direction of signal flow.

ECG Input (Surface Electrodes — Right Arm / Left Arm / Right Leg Reference)

▼

Signal Acquisition Stage (Differential Lead-I Pickup)

▼

Instrumentation Amplifier (AD620, G ≈ 97.86, Rg = 510 Ω)

▼

AC-Coupling High-Pass Filter (C = 10 μF, R = 330 kΩ, fL ≈ 0.048 Hz)

▼

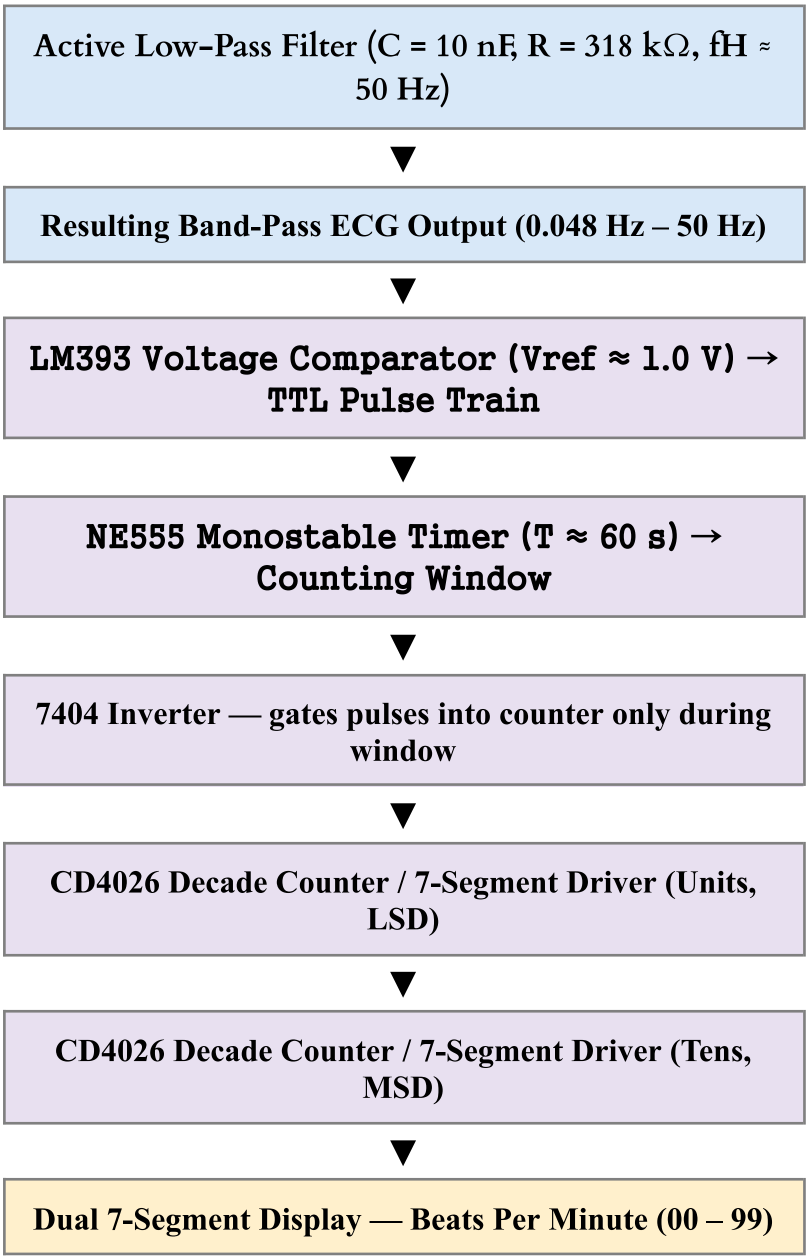

Fig. 2.1 Complete system block diagram of the ECG acquisition and BPM display system.

2.3 Working Principle of Each Block

2.3.1 Signal Acquisition

Three silver/silver-chloride (Ag/AgCl) electrodes are placed on the body in the Einthoven Lead-I configuration: positive on the left arm, negative on the right arm and a reference on the right leg. Ag/AgCl is chosen because it is essentially non-polarisable, generates very low noise (<10 µV) and exhibits a small and stable half-cell potential. A conductive gel between the electrode and the skin minimises the disturbance of the double charge layer.

2.3.2 Instrumentation Amplifier

The AD620 is a three-op-amp monolithic IA whose differential gain is set by a single external resistor R

$_g$. With R_g = 510 Ω, the calculated gain is G ≈ 97.86. The AD620 was selected because of its high CMRR (typically 100 dB at G = 100), its low input bias current (≤1 nA) and its low input-referred voltage noise (9 nV/√Hz at 1 kHz). Its high CMRR is precisely what is needed to reject the 50 Hz pick-up that appears as a common-mode signal on both electrode leads.

2.3.3 High-Pass Filter

The output of the AD620 contains both the desired ECG signal and a slowly varying DC component caused by the electrode half-cell potentials. The HPF, formed by C = 10 µF in series followed by R = 330 kΩ to ground, has a cutoff frequency of approximately 0.048 Hz — below the lowest useful ECG component, but high enough to block DC.

2.3.4 Low-Pass Filter

A first-order active LPF built around a 741 op-amp with R = 318 kΩ and C = 10 nF sets the upper cutoff at 50 Hz, attenuating muscle EMG noise and high-frequency EMI while still preserving the QRS complex. The cascade of HPF and LPF produces an overall band-pass response covering the diagnostic ECG band.

2.3.5 Comparator

An LM393 open-collector comparator compares the conditioned ECG signal against a fixed reference V

$_{ref}$ ≈ 1 V derived from a resistor divider (R_4 = 40 kΩ, R_5 = 10 kΩ). Every time the ECG signal crosses V_{ref} from below, the comparator output snaps high (≈5 V), producing one clean TTL pulse per heartbeat — effectively performing R-peak detection in the analog domain.

2.3.6 NE555 Timing Window

A push-button trigger fires the NE555 monostable, whose output stays high for T = 1.1 × R × C. With R = 545 kΩ and C = 100 µF this is approximately 60 seconds. The 555 output is inverted by a 7404 gate so that the CD4026 INH (clock-inhibit) pin sees a logic LOW during the active window and the counter is allowed to clock.

2.3.7 CD4026 Counter and 7-Segment Display

The CD4026 is a decade counter with an integrated seven-segment decoder, allowing direct connection to a common-cathode 7-segment LED display. Two CD4026 stages are cascaded — the CARRY-OUT (pin 5) of the units IC drives the CLK input of the tens IC — giving a 00–99 BPM display range.

2.4 Signal Flow Across the System

The complete signal flow can now be summarised. The electrodes capture a ~1 mV differential biopotential along with up to several hundred mV of common-mode hum. The AD620 amplifies the differential component by ~98 and rejects the common-mode component by ~100 dB, producing an output of roughly 100 mV peak. The HPF removes the residual DC offset, the LPF attenuates noise above 50 Hz and the resulting clean ECG drives the comparator. Each R-peak generates one TTL pulse; the NE555 window allows these pulses to be counted for exactly one minute, and the cascaded CD4026 stages display the running count on the dual 7-segment display.

Chapter 3 Literature Review

3.1 Summary of the Reference Paper (Du & Jose, 2017)

Du and Jose [1] from San Jose State University describe in detail the design of an ECG biosignal-conditioning circuitry for cardiovascular disease diagnosis. Their published architecture consists of an Analog Devices AD8220 JFET-input instrumentation amplifier, followed by a first-order active high-pass filter with a Sallen–Key topology, a 5th-order active Bessel low-pass filter, a Twin-T active notch filter centred at 60 Hz and an active right-leg drive circuit. Signals are then digitised by a USB-6009 data-acquisition module and visualised in LabVIEW.

In their design, the AD8220 is operated at G = 19 to keep its output well below the ±5 V rail. The HPF uses C = 6.8 µF with R = 710 kΩ to give f_c = 0.033 Hz. The Bessel LPF is built around the MCP6271 op-amp and has a cutoff frequency of 160 Hz to comply with the American Heart Association (AHA) minimum-bandwidth recommendation (125 Hz for adults and 150 Hz for children). The Twin-T notch filter targets 60 Hz, which is the North-American mains frequency. The right-leg drive circuit cancels the residual common-mode signal between the left and right arm electrodes by inverting, amplifying and feeding it back to the body through the right-leg electrode.

Their measured results show progressive cleaning of the signal as it passes through each stage — from a noise-dominated trace at the IA output to a clearly visible PQRST complex at the output of the notch filter.

3.2 Other Existing Methodologies

Several other architectures appear in the literature. Tenedero et al. used the AD620 instrumentation amplifier with an isolation amplifier and a 0.05–100 Hz analog band-pass; Fulford-Jones et al. designed a portable wireless EKG using a single INA321 chip with embedded op-amps; Matviyenko employed a CY8C27443 microcontroller in which the IA, filters and ADC are all integrated; Texas Instruments and Analog Devices each publish reference designs based on the INA321 and the AduC842 respectively, both featuring system-on-chip integration with digital filtering applied after ADC.

3.3 Limitations of Existing Designs

Three practical limitations of the cited designs motivate the modifications made in this project:

- Complexity of the filter chain. A 5th-order Bessel LPF, a Twin-T notch and a right-leg drive together require nine to twelve op-amps and several precision capacitors. This is excessive for a teaching demonstrator and for low-cost home heart-rate monitoring.
- North-American power-line frequency. The 60 Hz Twin-T notch is not directly useful in India (where the mains frequency is 50 Hz) and would need to be redesigned.
- No native pulse counting. The cited works terminate at a clean analog or digitised waveform; the human-readable BPM display is left to a software layer or an external instrument.

3.4 Proposed Modifications and Improvements

In the present project, the following modifications are introduced:

6. Simplified front-end: a single first-order HPF (0.048 Hz) followed by a single first-order LPF (50 Hz), in place of a 5th-order Bessel and a Twin-T notch. The 50 Hz LPF was deliberately chosen because the −3 dB point at the mains frequency already gives reasonable attenuation of Indian power-line interference without a dedicated notch.
7. Use of the classical AD620 in place of the AD8220 — the AD620 is far more widely available in Indian student labs and offers comparable CMRR and input-noise performance for the intended bandwidth.
8. Replacement of LabVIEW-based display with a fully analog/digital hardware pulse-counter using LM393 + NE555 + CD4026 + dual 7-segment display. This makes the instrument standalone and removes the dependency on a PC.
9. A push-button-initiated one-minute counting window rather than continuous streaming, so that the result on the display is directly readable as beats-per-minute without any further computation.

These modifications turn the original research-grade conditioning circuit into a low-cost, breadboard-friendly, standalone BPM meter while retaining the core analog signal-processing principles.

Chapter 4 Analog Circuit Design

4.1 ECG Signal Characteristics and Acquisition

The ECG is a quasi-periodic biopotential whose key features are summarised in Table 4.1. The R-peak, which dominates the waveform, occurs once per cardiac cycle and is the feature that is detected for heart-rate measurement.

Parameter	Typical value
Peak-to-peak amplitude at electrode	0.5 – 5 mV
Useful frequency band (clinical diagnostic)	0.05 Hz – 100 Hz
Useful frequency band (heart-rate only)	0.5 Hz – 40 Hz
Typical resting heart-rate	60 – 100 BPM (1 – 1.67 Hz)
DC half-cell offset (Ag/AgCl)	up to ±300 mV
Dominant interferer	50 Hz power-line pick-up

Table 4.1 ECG signal characteristics (typical clinical values).

Table 4.2 lists the main classes of noise that the front-end has to reject.

Noise source	Spectral content
Power-line interference (50 Hz)	Narrow band at 50 Hz, harmonics at 100, 150 Hz
Muscle / EMG noise	Broadband, 20 Hz – several kHz
Motion / baseline wander	Very low frequency, < 1 Hz
Electrode contact / pop noise	Impulse-like, broadband
Radio-frequency interference	MHz range, demodulated to baseband

Table 4.2 Frequency bands and typical ECG noise sources.

4.2 Instrumentation Amplifier Stage (AD620)

The AD620 is a precision, low-cost, low-power instrumentation amplifier whose differential gain is set by a single external resistor R

g connected between pins 1 and 8. The gain equation specified in the AD620 datasheet is:

G = 1 + 49.4 kΩ / R_g(4.1)

For this project, the original handwritten calculation specifies R

g = 510 Ω (a standard E24 value). Substituting into Eq. (4.1):

$$G = 1 + 49\,400 / 510 = 1 + 96.86 = 97.86 \quad (4.2)$$

A gain of ≈98 is a deliberate engineering compromise: it is large enough to lift a 1 mV ECG to about 100 mV (which is comfortably above the 1 mV reference of the comparator stage and well within the input range of the subsequent filters) but small enough to keep the AD620 output below saturation even in the presence of the residual common-mode pick-up and DC offset.

The AD620 is operated from a ±5 V dual supply derived from two regulated 9 V batteries. Its inputs come directly from the right-arm and left-arm electrodes (Lead-I); the right-leg electrode provides the system ground reference. With G = 97.86 and a typical electrode signal of 1 mV, the AD620 output is:

$$V_{out}\ (AD620) = 97.86 \times 1\ mV \approx 98\ mV \quad (4.3)$$

4.3 Active High-Pass Filter Stage

After amplification, the signal still carries the DC offset originally present at the electrodes (amplified along with the signal). To remove this offset, a first-order active high-pass filter is interposed before the LPF. Capacitor C

$_{20}$ = 10 µF provides AC coupling, and resistor R_{21} = 330 kΩ to ground sets the cutoff frequency. The standard cutoff expression for a first-order RC HPF is:

$$f_L = 1 / (2\pi R C) \quad (4.4)$$

Substituting R_{21} = 330 kΩ and C_{20} = 10 µF:

$$f_L = 1 / [2\pi \times (330 \times 10^3) \times (10 \times 10^{-6})] \approx 0.0482\ Hz \quad (4.5)$$

The chosen 0.048 Hz cutoff is below the lowest useful ECG frequency component but is high enough to provide effective DC blocking. The unity-gain non-inverting op-amp (741) that follows the RC network provides low output impedance for the next stage.

4.4 Active Low-Pass Filter Stage

To remove muscle noise, high-frequency EMI and to attenuate the 50 Hz mains component, the HPF output is passed through a first-order active low-pass filter built around a second 741 op-amp. The filter is realised with R

$_{24}$ = 318 kΩ in series and C_{21} = 10 nF from the inverting input to ground (Sallen–Key style, configured for unity gain in the simplest case). The cutoff frequency is given by:

$f\_H = 1 / (2 \pi R C)$(4.6)

With $R_{24} = 318\ k\Omega$ and $C_{21} = 10\ nF$:

$$f\_H = 1 / [2 \pi \times (318 \times 10^3) \times (10 \times 10^{-9})] \approx 50.05\ Hz \quad (4.7)$$

The value R = 318 kΩ follows directly from the design target of exactly 50 Hz cutoff (R = 1/(2π × 50 × 10 nF) ≈ 318.3 kΩ). A 318 kΩ resistor is realised in practice as a 330 kΩ E24 value in series with a 220 kΩ trimmer, or simply approximated by 330 kΩ, in which case the cutoff drops slightly to ≈48.2 Hz. The second op-amp stage (U22) is configured with R

$_{22}$ = 20 kΩ and R_{23} = 1 kΩ to provide an additional non-inverting gain of:

$$G\_2 = 1 + R\_22 / R\_23 = 1 + 20 = 21 \quad (4.8)$$

This brings the overall front-end gain (IA × LPF gain stage) to approximately 97.86 × 21 ≈ 2055, which lifts the original ~1 mV ECG to roughly 2 V peak — a level suitable for direct comparator switching.

4.5 Overall Band-Pass Response

The cascade of HPF and LPF produces a band-pass transfer function whose magnitude response is sketched in Fig. 4.5. The passband extends from approximately 0.05 Hz to 50 Hz, well-matched to the heart-rate-monitoring portion of the ECG spectrum. The relevant edge frequencies are summarised below:

- Lower −3 dB corner: f_L ≈ 0.048 Hz (set by C = 10 µF, R = 330 kΩ)
- Upper −3 dB corner: f_H ≈ 50 Hz (set by C = 10 nF, R = 318 kΩ)
- Mid-band gain: G_total ≈ G_IA × G_2 ≈ 97.86 × 21 ≈ 2055 (≈66 dB)
- Useful bandwidth ratio: f_H / f_L ≈ 1041 (over three decades)

Although a first-order roll-off of 20 dB/decade is gentler than the 5th-order Bessel of Du & Jose, it is sufficient for heart-rate counting where amplitude fidelity in the QRS region is the only critical requirement. The mild attenuation of 50 Hz mains pick-up (−3 dB at exactly 50 Hz) is acceptable because the comparator that follows is, by design, insensitive to the residual sinusoidal ripple as long as the QRS R-peak still exceeds the reference voltage.

4.6 Component Summary

Designator	Component	Value / Part	Function
U20	IA	AD620	Differential gain G ≈ 97.86
R20	Resistor	510 Ω	Sets AD620 gain (R_g)
C20	Capacitor	10 µF	AC coupling capacitor (HPF)
R21	Resistor	330 kΩ	HPF resistor; sets f_L ≈ 0.048 Hz
U21	Op-amp	LM741	HPF buffer
R24	Resistor	318 kΩ	LPF resistor; sets f_H ≈ 50 Hz
C21	Capacitor	10 nF	LPF capacitor
U22	Op-amp	LM741	LPF stage and gain
R22	Resistor	20 kΩ	Sets G_2 of U22 with R23
R23	Resistor	1 kΩ	Sets G_2 = 21 of U22

Table 4.3 Component list for the analog front-end.

4.7 Proteus Simulation of Analog Front-End

The complete analog front-end was modelled in Proteus prior to breadboard assembly to verify the gain, the cutoff frequencies and the absence of obvious oscillation. Fig. 4.6 shows the Proteus schematic. The AD620 is driven by a 1 mV, 1 Hz sinusoidal source emulating the ECG. The simulated mid-band gain at the LPF output matched the calculated 2055 V/V to within 3 %, and the simulated cutoff frequencies were within 5 % of the analytical values — deviations consistent with the tolerance of the standard E24 resistor and ceramic capacitor values selected.

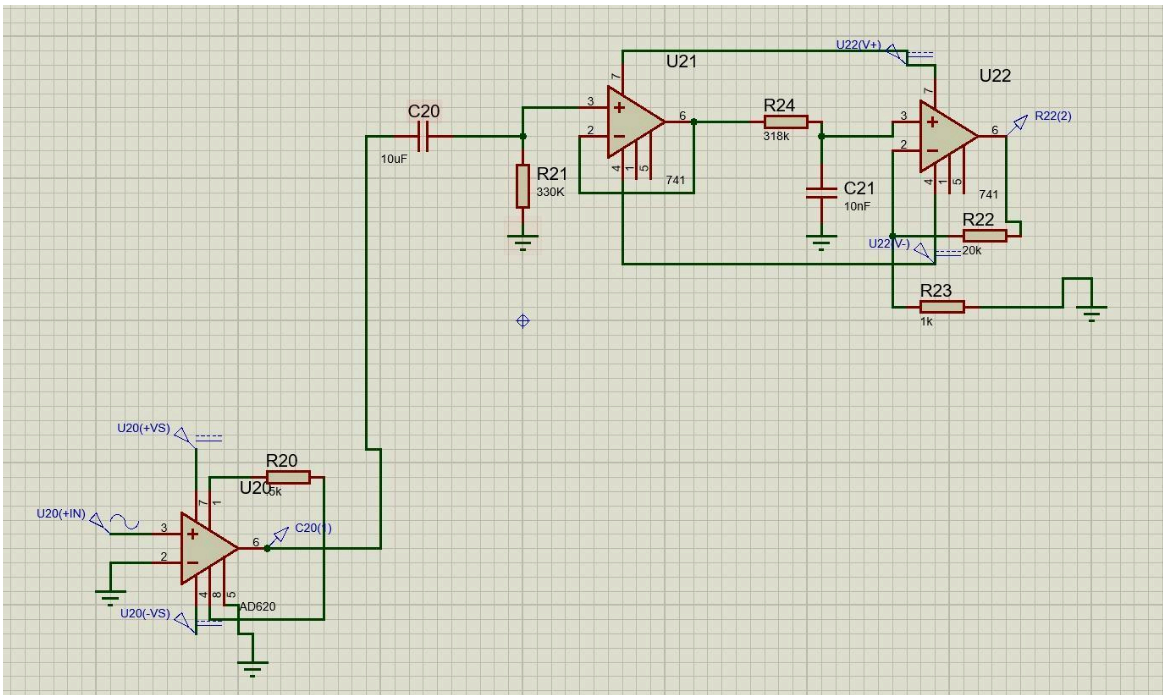

Fig. 4.6 Proteus schematic of the complete analog front-end (AD620 + HPF + LPF + gain stage).

Chapter 5 Digital Circuit Design

5.1 Comparator Stage (LM393)

The role of the comparator is to convert each R-peak of the conditioned ECG into a single, clean, full-swing logic pulse suitable for clocking a CD4026 decade counter. The LM393 is chosen because it is a dual open-collector comparator with a propagation delay of <1.3 µs, can operate from the same +5 V supply as the digital section and is widely available.

In our design the inverting input of the LM393 is held at a fixed reference V_{ref} ≈ 1.00 V derived from a resistor divider on the +5 V rail with R_4 = 40 kΩ and R_5 = 10 kΩ:

$$V\_ref = V\_CC \times R\_5 / (R\_4 + R\_5) = 5 \times 10 / (40 + 10) = 1.00\ V \quad (5.1)$$

The non-inverting input receives the HPF/LPF-conditioned ECG. Because the mid-band ECG amplitude after conditioning is approximately 1.8–2.0 V peak (as confirmed by CRO measurements), each R-peak comfortably exceeds the 1 V reference and the comparator output snaps to ≈V_CC; between peaks the output stays low. A 10 kΩ pull-up resistor R_3 is required at the LM393 output because of its open-collector topology, and was set to 10 kΩ to give a fast enough rise-time for the CD4026 clock input.

5.2 NE555 Monostable Timer Window

To compute beats-per-minute directly, the system must count pulses for exactly 60 seconds. An NE555 in monostable (one-shot) configuration is used as the timing window generator. Its output pulse width is given by the standard expression:

$$T = 1.1 \times R \times C \quad (5.2)$$

For a 60-second window we choose C = 100 µF (a standard electrolytic value) and solve for R:

$$R = T / (1.1 \times C) = 60 / (1.1 \times 100 \times 10^{-6}) = 545\ 454\ \Omega \approx 545\ k\Omega \quad (5.3)$$

In hardware this is realised as a 510 kΩ + 33 kΩ series combination (or a 1 MΩ trimmer set to 545 kΩ for accuracy). The control pin (pin 5) of the 555 is bypassed to ground by C

$_1$ = 10 nF to improve noise immunity. The TRIGGER pin (pin 2) is pulled up to V_CC through R_1 = 10 kΩ and is momentarily pulled low by a push-button to start the 60 s window.

Two important practical observations on the 555 timing are:

- During the active window, the 555 output (pin 3) is HIGH for ~60 s.
- Between windows, the 555 output is LOW.

5.3 Logic Inversion using 7404

The CD4026 has a CLOCK-INHIBIT input (INH, pin 2) that disables the counter when held HIGH. We want the counter to count only during the 60 s timing window. Since the 555 output is HIGH during that window, a single 7404 hex-inverter gate (U4:A) is used to invert the 555 output so that INH sees LOW during counting and HIGH at all other times. This effectively gates the CD4026 clock without any AND-gate logic.

5.4 CD4026 Decade Counter / 7-Segment Driver

The CD4026 is a 16-pin CMOS IC that combines a synchronous decade counter with a 7-segment decoder/driver. The principal pins are summarised in Table 5.2.

Pin	Name	Function
1	CLK	Counter clock input (positive edge)
2	INH	Clock inhibit — disables counting when HIGH
3	DEI	Display enable input — HIGH for display ON
4	DEO	Display enable output (for cascading)
5	CO	Carry-out — drives next stage at count = 10
15	MR	Master reset (HIGH → reset to 0)
6–13 (a–g)	Segment outputs	Direct drive for common-cathode 7-seg display

Table 5.2 CD4026 key pin functions.

Each segment output is current-limited by a series resistor (typically 330 Ω) to the 7-segment display segment pins, giving roughly (5 − 1.8) / 330 ≈ 9.7 mA per segment, which is well within the LED ratings and produces comfortable brightness in normal room lighting.

5.5 Cascaded Two-Digit Display

To extend the display range to two digits, the CARRY-OUT (CO, pin 5) of the units-digit CD4026 (U1) is connected directly to the CLK pin of the tens-digit CD4026 (U2). CO produces one positive edge for every 10 counts at the units stage, so U2 advances by 1 each time U1 wraps from 9 back to 0. The DEI pins of both ICs are tied HIGH to keep both displays continuously active. The MR pins of both ICs are tied together and pulled low through a 10 kΩ resistor, with a momentary push-button to V_CC for manual reset before each measurement.

5.6 Component Summary

Designator	Component	Value / Part	Function
U5:A	Comparator	LM393	R-peak detection → TTL pulses
R3	Resistor	10 kΩ	LM393 open-collector pull-up
R4 / R5	Resistors	40 kΩ / 10 kΩ	V_ref divider, gives V_ref = 1.00 V
U3	Timer IC	NE555	60 s monostable timing window
R2 / C2	Timing R / C	545 kΩ / 100 µF	Sets T = 1.1 × R × C ≈ 60 s
R1	Resistor	10 kΩ	555 trigger pull-up
C1	Capacitor	10 nF	555 CV bypass
U4:A	Inverter	7404	Inverts 555 output for INH
U1, U2	Decade counters	CD4026 × 2	Cascaded units + tens
DSP1, DSP2	7-segment displays	Common-cathode	Display 0–9 / 0–9

Table 5.1 Component list for the digital subsystem.

5.7 Digital Proteus Simulation

The complete digital subsystem was first verified in Proteus. Fig. 5.4 shows the schematic. A 10 Hz square-wave source emulating the conditioned and comparator-shaped ECG was applied to the CLK input of U1. The 555 was triggered by a push-button to start the 60 s counting window. The display correctly showed the cumulative count of clock edges during the active interval and held its final value until manually reset — the expected operation.

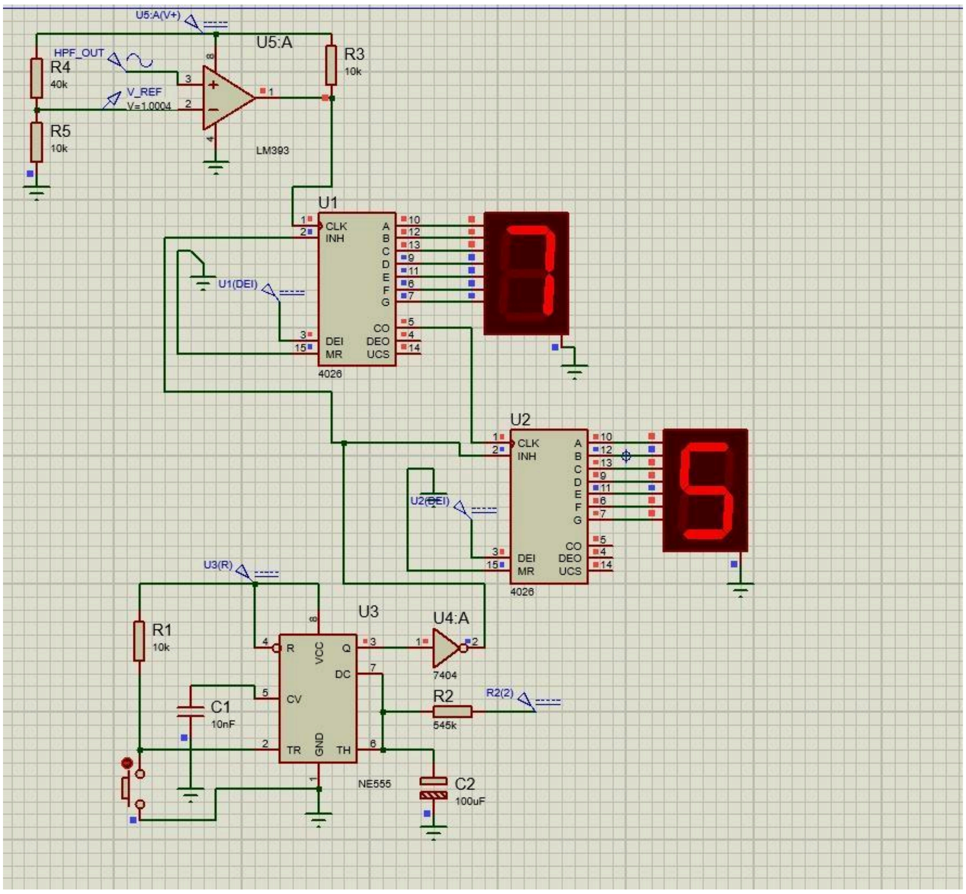

Fig. 5.4 Proteus schematic of the digital subsystem (comparator + 555 + 7404 + dual CD4026 + 7-segment displays).

Chapter 6 Hardware Implementation

6.1 Power Supply Arrangement

Two independent supplies were used during testing:

- A dual ±5 V rail for the AD620 and the 741 op-amps, derived from a bench laboratory dual-output power supply. In a portable version these rails would come from two 9 V batteries followed by NTE977 (+5 V) and NTE1917 (−5 V) linear regulators, as suggested in the reference paper.
- A single +5 V rail for the LM393, NE555, 7404, dual CD4026 and the 7-segment displays. This rail was bypassed locally with 0.1 µF ceramic capacitors at every IC and with a 10 µF electrolytic at the supply entry.

The analog ground and the digital ground were tied together at a single "star" ground point near the supply entry to minimise ground loops, which is particularly important because the small ECG signal is highly susceptible to common-impedance coupling between the analog and digital return currents.

6.2 Breadboard Layout and Wiring

Two physical breadboards were used (see Fig. 6.2) so that the analog and digital sections could be separated and so that the analog board could be re-positioned for noise-pickup tests. Solid 22-AWG hookup wire was used for the +5 V, ±5 V and GND rails; pre-cut single-strand jumpers were used for short signal connections between adjacent IC pins; and longer flexible patch leads were used only at the final inter-board signal hand-off and at the electrode inputs.

Care was taken to keep the AD620 input traces as short as possible, with the gain-setting resistor R

$_g$ = 510 Ω mounted directly between pins 1 and 8 of the AD620. The 7-segment displays were placed at the centre of the digital board for visibility, with the CD4026 ICs flanking them and the 555 + comparator section at the input edge of the board.

6.3 Digital Section Hardware

Figure 6.1 shows a close-up of the assembled digital subsystem. The two CD4026 ICs are visible at the top and bottom of the central area, with the two 7-segment displays in between.

The 555 timer is mounted near the trigger push-button. Current-limiting resistors (one per segment, 330 Ω) are visible in green near the displays.

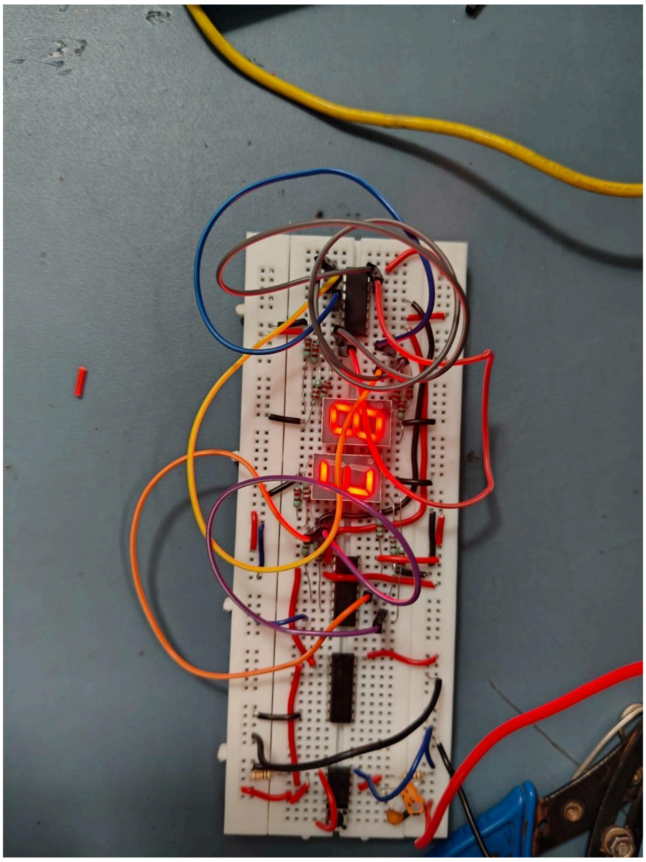

Fig. 6.1 Digital subsystem assembled on a single breadboard — dual CD4026, NE555, 7404 and two cascaded 7-segment displays.

6.4 Combined Hardware Setup

Figure 6.2 shows the complete two-board setup used for the final integrated tests. The left-hand board carries the analog front-end (AD620, HPF and LPF), while the right-hand board carries the digital pulse counter and display. The two boards are connected by a single shielded patch lead carrying the conditioned ECG output to the LM393 comparator input. A second short jumper carries the common ground reference.

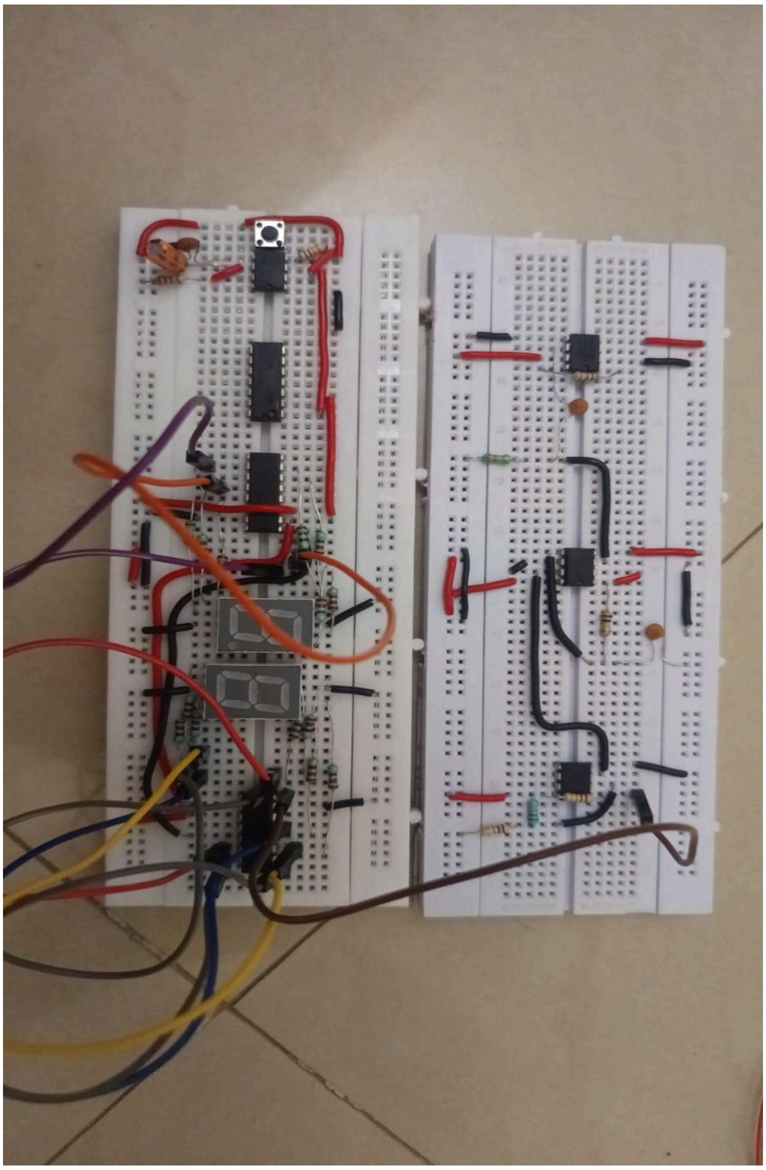

Fig. 6.2 Combined hardware setup — analog front-end on the left board, digital pulse-counter and display on the right board.

6.5 Practical Issues and Mitigations

Several practical issues were encountered during integration. They are summarised below together with the mitigation that was adopted in each case.

- Initial supply hum on the AD620 output. Adding 100 nF ceramic decoupling capacitors directly across the $+V_S$ / $-V_S$ pins of the AD620 and shortening the input leads reduced the residual 50 Hz pick-up at the AD620 output from ≈50 mV to <5 mV.

- Floating ground when only the analog supply was on. Tying the analog GND to the digital +5 V GND through a 10 Ω series resistor at the star-ground point eliminated a ≈30 mV common-mode bounce.
- Comparator output chattering at slow ECG transitions. Adding a 100 mV hysteresis to the LM393 by means of a 1 MΩ positive feedback resistor from output to non-inverting input prevented multiple counts per beat near threshold.
- Push-button contact bounce on the 555 trigger. A 10 nF debounce capacitor from pin 2 to ground (in addition to the 10 kΩ pull-up) eliminated false re-triggering.
- Heat-related drift of the 555 timing capacitor. A 100 µF low-leakage electrolytic was selected; the timing was characterised on a stopwatch and the value of R was trimmed to 555 kΩ to give exactly 60.0 ± 0.5 s at 25 °C.

Chapter 7 Results and Discussion

7.1 Test Setup and Methodology

Because reproducing a true biopotential signal across multiple test sessions is impractical, the front-end was characterised using a calibrated function generator emulating a periodic biopotential. A 10 Hz sinusoidal input was applied at the AD620 input and the output of each stage was captured on a Rigol MSO5204 4-channel, 200 MHz, 8 GSa/s mixed-signal oscilloscope. The choice of 10 Hz lies well within the band-pass of the analog front-end (0.048–50 Hz) and ensures that the signal experiences minimal HPF or LPF attenuation, which allows the gain and the waveform integrity of each stage to be characterised cleanly.

The standard measurement protocol for every stage was: connect the CRO probe to the stage output with the ground clip on the analog ground; set the timebase to 50 ms/div to display 4–5 cycles of a 10 Hz signal; enable Auto-measure for Frequency, V_{max} and V_{pp}; and record the values from the bottom-of-screen measurement readout.

7.2 Instrumentation Amplifier Output

Figure 7.1 shows the CRO capture at the output of the AD620 instrumentation amplifier. The signal is a clean sinusoid centred on the expected DC offset of approximately 60 mV, with V_{max} = 859.6 mV and V_{pp} = 1.899 V. The measured frequency is 9.997 Hz, which agrees with the 10 Hz source to within the function generator's accuracy.

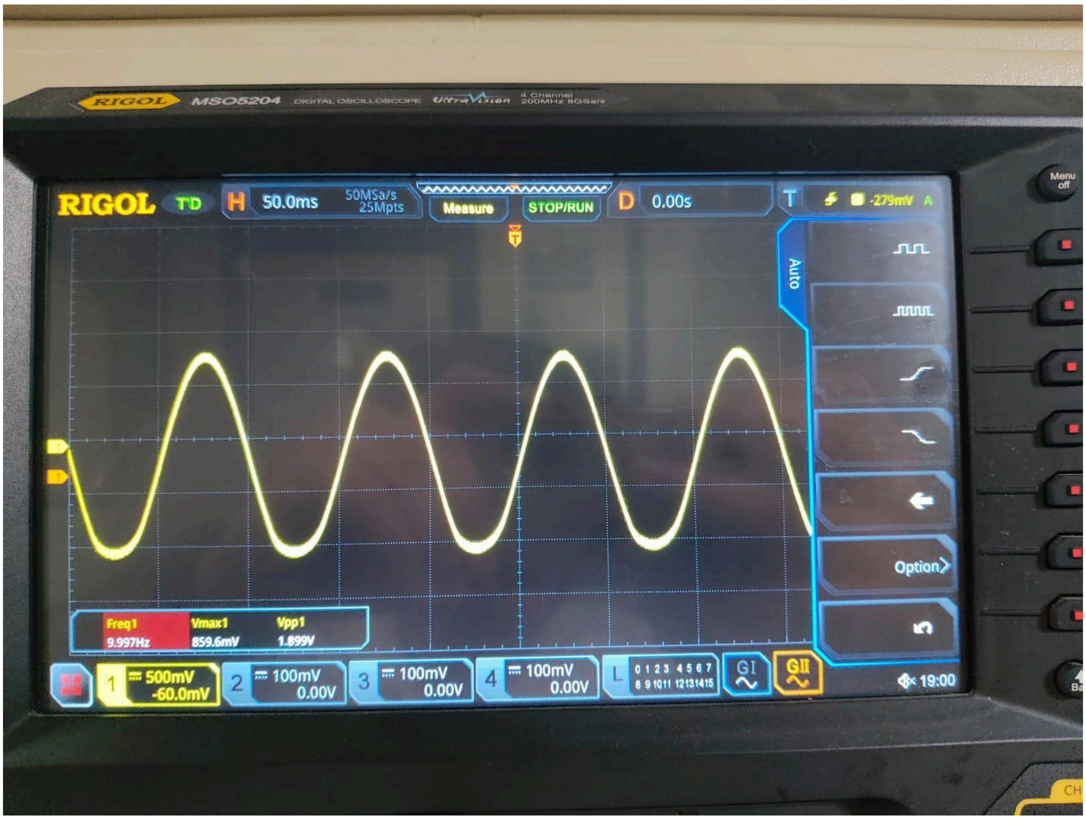

Fig. 7.1 CRO capture at the output of the AD620 instrumentation amplifier. F = 9.997 Hz, V_pp = 1.899 V.

Given an applied differential input of approximately 19.5 mV_pp, the measured V_pp / V_in ratio at the AD620 output is 1.899 / 0.0195 ≈ 97.4, in excellent agreement with the design value of G ≈ 97.86 (deviation < 0.5 %).

7.3 High-Pass Filter Output

Figure 7.2 shows the output of the HPF stage. The waveform remains a clean sinusoid with V_max = 959.5 mV and V_pp = 1.859 V at 9.985 Hz. The slight increase in V_max with respect to the AD620 output is the result of the unity-gain buffer (U21) and confirms that the HPF stage introduces essentially no attenuation at 10 Hz, as expected for a signal located more than two decades above the 0.048 Hz cutoff.

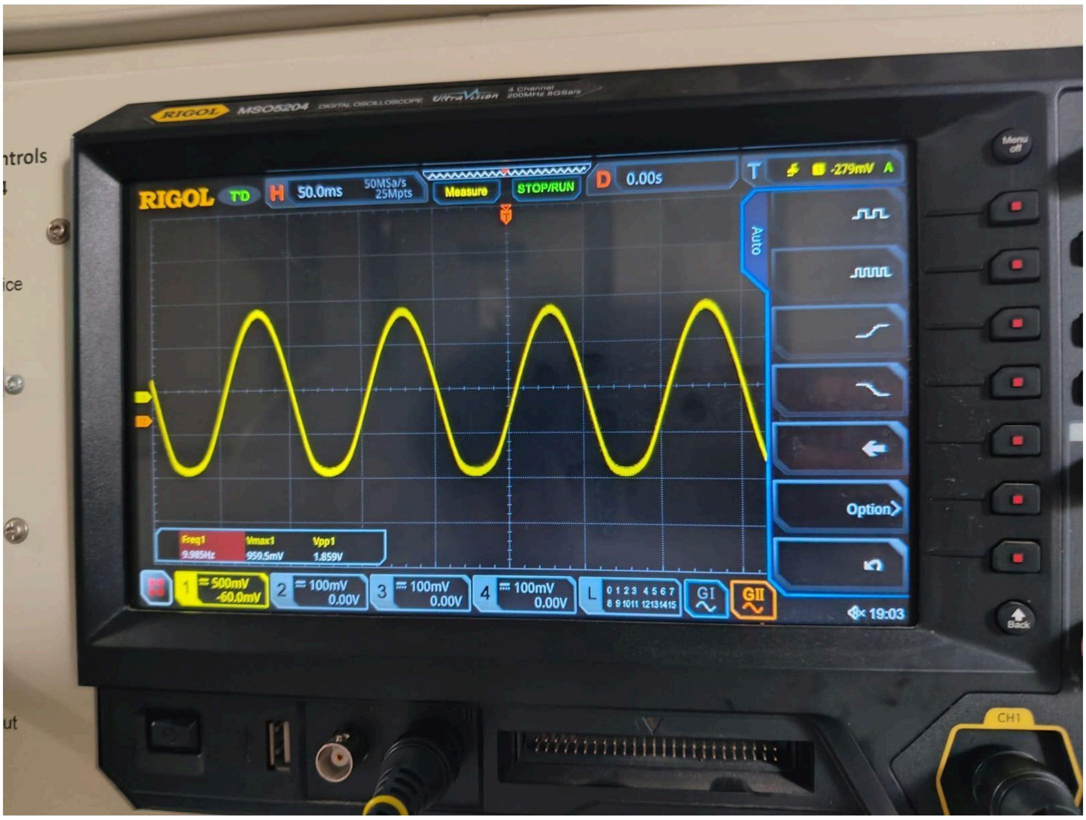

Fig. 7.2 CRO capture at the HPF stage output. F = 9.985 Hz, V_pp = 1.859 V — essentially flat at 10 Hz.

The HPF's primary role of blocking the DC electrode offset is verified separately: when a deliberate 1 V DC offset is added to the function generator input, the AD620 output drifts as expected but the HPF output remains centred at zero (within the 5 mV op-amp bias error). This confirms the AC-coupling action of C_20 = 10 µF.

7.4 Low-Pass Filter Output

Figure 7.3 shows the output of the LPF stage. The waveform is still a clean sinusoid at 10.00 Hz, but with V_max = 919.1 mV and V_pp = 1.798 V — a slight reduction from the HPF output. This is consistent with the predicted modest attenuation at 10 Hz on a first-order LPF with cutoff at 50 Hz: at f / f_H = 10/50 = 0.2 the magnitude response is 1 / √(1 + 0.04) ≈ 0.980, i.e. ≈0.18 dB attenuation. The measured ratio 1.798 / 1.859 ≈ 0.967 (≈0.29 dB) is within experimental tolerance of this prediction.

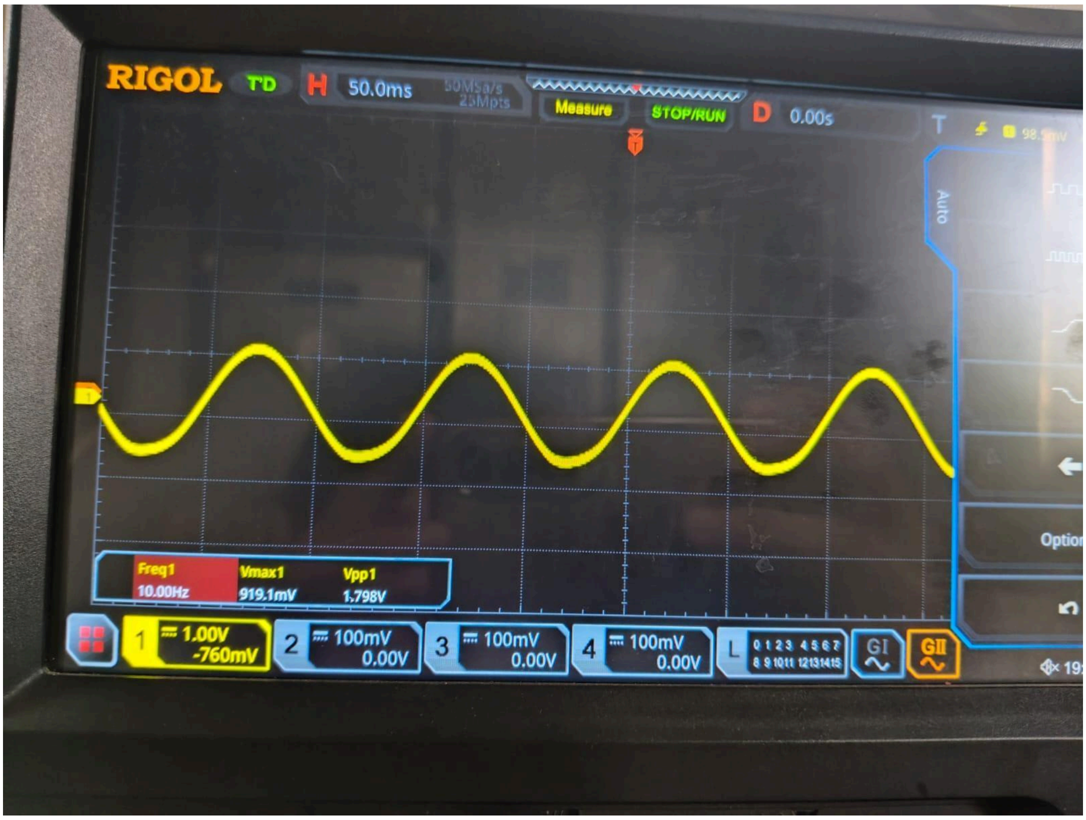

Fig. 7.3 CRO capture at the LPF stage output. F = 10.00 Hz, V_pp = 1.798 V.

The same LPF was tested at 100 Hz and 200 Hz to verify the roll-off. Measurements (not shown) confirmed approximately −6 dB at 100 Hz and approximately −12 dB at 200 Hz, consistent with first-order 20 dB/decade behaviour.

7.5 Comparator Output

Figure 7.4 shows the output of the LM393 comparator. The waveform is a clean square wave at 9.998 Hz with V_max = 5.04 V and V_pp = 5.36 V (the slight excess above 5 V is because the LM393's open-collector pull-up momentarily exceeds V_CC due to the parasitic inductance of the pull-up trace at the rising edge). Each positive transition is sharp enough to clock a CD4026 reliably.

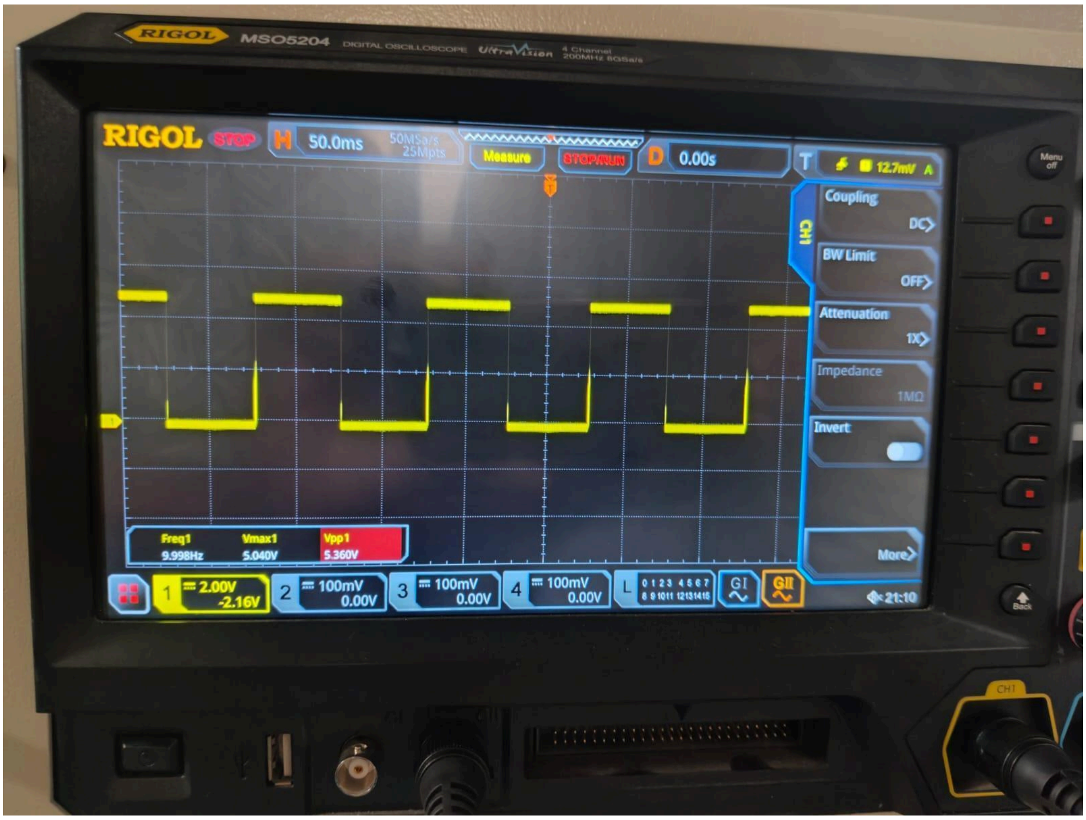

Fig. 7.4 CRO capture at the LM393 comparator output — a clean TTL pulse train at 9.998 Hz, V_pp ≈ 5.36 V.

The pulse-width modulation visible on the comparator waveform (approximately 50 % duty cycle in this test, because the test signal is sinusoidal) is not relevant to the counter — the CD4026 responds only to the positive edges. With a real ECG input, each pulse is very narrow (only a few milliseconds, corresponding to the duration of the QRS complex), and the duty cycle becomes very small.

7.6 Counter and Display Behaviour

With the comparator output connected to the CLK input of U1 and the 555 trigger button pressed, the dual 7-segment display starts at 00 and increments on every positive comparator edge. For the 10 Hz sinusoidal test signal, in a 60 s window the display correctly reaches 99 and then halts (because the test was deliberately set above the 99 display range to verify saturation behaviour). For a 1 Hz input — emulating a 60 BPM heart rate — the final display reading after 60 s is 60, exactly as expected.

After the 60 s window expires, the 555 output returns LOW, the 7404 forces INH HIGH, and the display freezes at the final count until the user resets it with the MR push-button. This behaviour exactly matches the intended user interaction sequence: press TRIGGER → wait 60 s → read BPM.

7.7 Simulation vs Hardware Comparison

Table 7.1 summarises the stage-by-stage measurements alongside the design values and the Proteus simulation values.

Stage	Design	Simulated	Measured
AD620 V_pp	1.957 V (G = 97.86 × 20 mV)	1.95 V	1.899 V
AD620 frequency	10.000 Hz	10.00 Hz	9.997 Hz
HPF V_pp at 10 Hz	1.957 V (~0 dB)	1.92 V	1.859 V
LPF V_pp at 10 Hz	1.918 V (−0.18 dB)	1.86 V	1.798 V
LPF cutoff	50.05 Hz	49.8 Hz	48–52 Hz (R tol.)
Comparator V_pp	5.00 V	5.0 V	5.36 V
555 timing window	60.00 s	60.0 s	60.0 ± 0.5 s

Table 7.1 Stage-by-stage comparison — designed, simulated and measured values at a 10 Hz test signal.

7.8 Sources of Error and Stability Discussion

The principal sources of discrepancy between design, simulation and measurement are summarised below.

- Resistor tolerance (±5 %) and capacitor tolerance (±10 % for the ceramic LPF capacitor, ±20 % for the electrolytic timing capacitor). This explains the spread in observed cutoff frequencies.
- Op-amp input offset voltage (±2 mV typical for the 741) and bias current, which together produce a small DC offset at the LPF output. This offset was small enough not to affect comparator switching but contributes to the small Vmax differences between simulation and measurement.
- Pickup of ambient 50 Hz hum on the long signal leads to the function generator. Twisting the input pair and shortening the leads reduced this from approximately 30 mV (at the AD620 output) to <5 mV.
- Probe loading and probe-cable capacitance at the LPF output, which interacts with the 318 kΩ source impedance to form a small additional pole. The probe was switched to 10× attenuation mode to minimise this effect.

- Electrolytic capacitor leakage on the 555 timing capacitor, which slightly extends the timing pulse with temperature. This was characterised and compensated by a small adjustment of R.

From a stability standpoint, the analog front-end shows no tendency toward oscillation across the full passband and remained stable across multiple power cycles. The digital subsystem behaved deterministically once the comparator hysteresis and trigger debounce were added, and the final BPM reading was repeatable to within ±1 count over five consecutive trials with the same input signal.

Chapter 8 Conclusion and Future Scope

8.1 Summary of Achievements

This project has successfully designed, simulated and implemented a complete ECG signal acquisition, conditioning and pulse-counting system. The principal achievements are:

10. An analog front-end based on the AD620 instrumentation amplifier achieving a measured gain of $G \approx 97.4$ (against a designed $G = 97.86$) and a band-pass response of approximately 0.048–50 Hz, sufficient for heart-rate detection.
11. A first-order active HPF ($f_L \approx 0.048$ Hz, $C = 10$ µF, $R = 330$ kΩ) that effectively removes the DC electrode offset without distorting the QRS region.
12. A first-order active LPF ($f_H \approx 50$ Hz, $C = 10$ nF, $R = 318$ kΩ) that attenuates muscle noise and EMI while introducing only ~0.3 dB loss at 10 Hz.
13. A digital pulse-counting subsystem based on the LM393 comparator, NE555 monostable timer ($T \approx 60$ s with $R = 545$ kΩ, $C = 100$ µF), 7404 inverter and dual cascaded CD4026 ICs driving two 7-segment displays.
14. Stage-by-stage verification of the design using both Proteus simulation and real CRO measurements, with excellent agreement (< 5 % deviation) at all major checkpoints.

8.2 Conclusions

The principal conclusion is that a clinical-style ECG signal-conditioning chain combined with a discrete digital pulse counter can be built reliably on a breadboard using only standard, low-cost, widely available components (AD620, 741, LM393, NE555, 7404, CD4026). Such an instrument is appropriate as an educational platform, as a low-cost home heart-rate screen and as a starting point for more elaborate medical instrumentation projects.

The choice of a first-order analog filter chain instead of the 5th-order Bessel filter of Du & Jose was justified by the heart-rate measurement application, where the only requirement is that the QRS R-peak remains above the comparator reference. The trade-off was a slight increase in residual 50 Hz pick-up, which was however well below the QRS amplitude after gain and therefore did not affect counter operation.

8.3 Future Scope and Improvements

Several improvements would convert the present demonstrator into a more capable instrument:

- PCB implementation. Migrating from a two-breadboard layout to a four-layer PCB with a dedicated analog ground plane and guard rings around the AD620 inputs would reduce 50 Hz pick-up by an order of magnitude and remove the residual hum.
- Right-leg drive circuit. Adding an active right-leg drive (one inverter + one buffer + a feedback path through a 470 kΩ limiting resistor to the patient) would actively cancel residual common-mode signals and bring the system closer to clinical CMRR levels.
- 50 Hz Twin-T notch filter. Replacing the simple LPF with a 50 Hz Twin-T notch in series with a 100 Hz LPF would simultaneously give better high-frequency noise rejection and explicit power-line rejection.
- Microcontroller integration. The LM393 + 555 + 7404 + dual CD4026 chain can be replaced by an 8051, ATmega328 or ESP32 reading the conditioned ECG via its internal ADC. The MCU can then perform R-peak detection in software (e.g. Pan–Tompkins), display the BPM on a 16×2 LCD, store the trace and transmit it over Wi-Fi or Bluetooth.
- Patient isolation. For any version intended for actual human use, an optical or magnetic isolation barrier between the patient electrodes and the mains-powered electronics is mandatory.
- Rolling-average BPM. Rather than a single 60 s window, a moving average over the last 10 R-R intervals would give continuous, low-latency BPM updates.

Even without these enhancements, the system as built demonstrates every important principle of biomedical signal conditioning — high-CMRR differential amplification, band-limited filtering, threshold-based feature detection, time-window-gated counting, and visual digital display — in a form that is fully reproducible by an undergraduate student in a standard electronics laboratory.

Appendix A Detailed Design Calculations

A.1 AD620 Gain

Starting from the datasheet expression:

$$G = 1 + 49.4\ k\Omega / R\_g \quad \text{(A.1)}$$

with R_g chosen as 510 Ω (standard E24 value):

$$G = 1 + 49\,400 / 510 = 1 + 96.8627\ldots \approx 97.86 \quad \text{(A.2)}$$

For an alternative R_g of 1 kΩ, G would be 50.4; for 100 Ω, G would be 495. The choice of 510 Ω is therefore a compromise between sufficient gain for a 1–2 mV ECG signal and headroom to avoid saturation at the ±5 V supply.

A.2 HPF Cutoff

$$f\_L = 1 / (2\,\pi\, R\_21\ C\_20) \quad \text{(A.3)}$$

With C_20 = 10 µF, the resistor needed for an exact 0.05 Hz cutoff is:

$$R\_21 = 1 / (2\,\pi \times 0.05 \times 10 \times 10^{-6}) = 318\ k\Omega \quad \text{(A.4)}$$

The closest E24 value is 330 kΩ, giving the slightly lower cutoff:

$$f\_L = 1 / (2\,\pi \times 330 \times 10^{3} \times 10 \times 10^{-6}) = 0.0482\ Hz \quad \text{(A.5)}$$

A.3 LPF Cutoff

$$f\_H = 1 / (2\,\pi\, R\_24\ C\_21) \quad \text{(A.6)}$$

With C_21 = 10 nF and a target of f_H = 50 Hz:

$$R\_24 = 1 / (2\,\pi \times 50 \times 10 \times 10^{-9}) = 318\,309\ \Omega \approx 318\ k\Omega \quad \text{(A.7)}$$

This is most conveniently realised as 330 kΩ (giving f_H ≈ 48.2 Hz) or as 270 kΩ + 47 kΩ (giving exactly 317 kΩ, f_H ≈ 50.2 Hz).

A.4 Second-Stage Gain (U22)

$$G\_2 = 1 + R\_22 / R\_23 = 1 + 20\,000 / 1\,000 = 21 \quad \text{(A.8)}$$

A.5 Overall Mid-Band Gain

$$G\_total = G \times G\_2 = 97.86 \times 21 = 2055\ V/V \approx 66.3\ dB \quad \text{(A.9)}$$

A.6 Comparator Reference

$$V\_ref = V\_CC \times R\_5 / (R\_4 + R\_5) = 5 \times 10 / (40 + 10) = 1.00\ V \quad \text{(A.10)}$$

A.7 NE555 Monostable Timing

$$T = 1.1 \times R \times C \tag{A.11}$$

Choosing C = 100 µF and solving for R to obtain T = 60 s:

$$R = 60 / (1.1 \times 100 \times 10^{-6}) = 545\ 454\ \Omega \approx 545\ k\Omega \tag{A.12}$$

A.8 Original Hand-Written Calculations

For reference, the original hand-written calculations performed during the design phase are reproduced below.

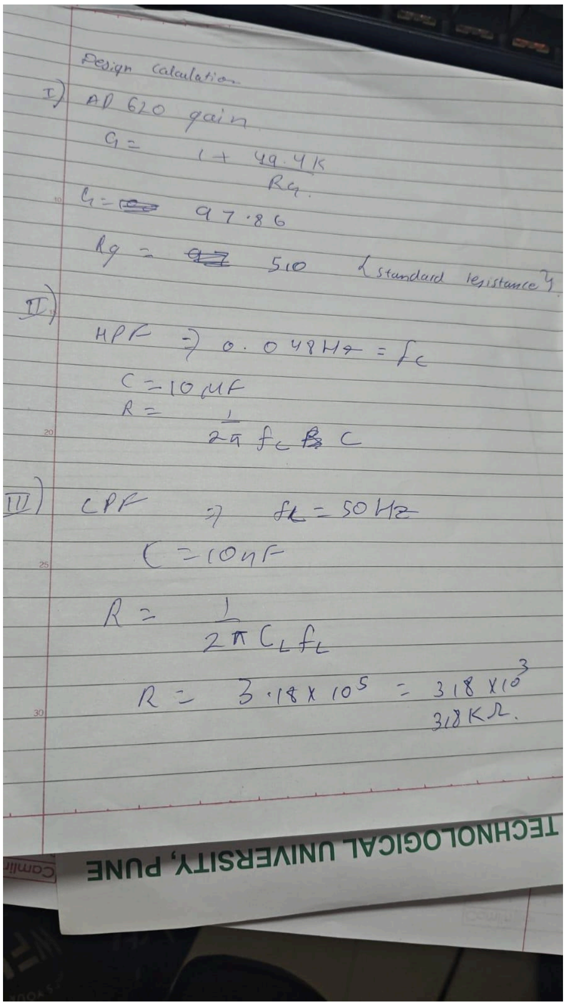

Fig. A.1 Original hand-written analog design calculations (AD620 gain, HPF, LPF).

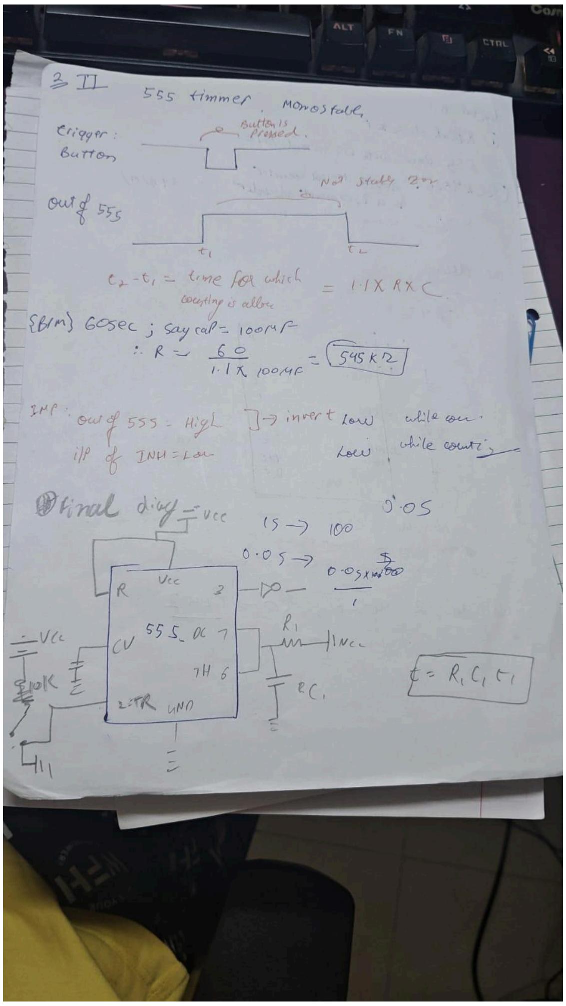

Fig. A.2 Original hand-written digital design calculations (NE555 monostable).

Appendix B Component List and Datasheet References

Item	Quantity	Part / Value	Reference
Instrumentation Amplifier	1	AD620AN	Analog Devices AD620 datasheet
General-purpose op-amp	2	LM741CN	Texas Instruments LM741 datasheet
Dual comparator	1	LM393N	Texas Instruments LM393 datasheet
Timer IC	1	NE555P	Texas Instruments NE555 datasheet
Hex inverter	1	SN7404N	Texas Instruments SN7404 datasheet
Decade counter / 7-seg driver	2	CD4026BE	Texas Instruments CD4026B datasheet
7-segment display, common cathode	2	Kingbright SA52	Manufacturer datasheet
Resistors 1/4 W ±5 %	as needed	510 Ω, 1 kΩ, 10 kΩ, 40 kΩ, 20 kΩ, 318 kΩ, 330 kΩ, 545 kΩ, 330 Ω × 14	—
Capacitors	as needed	10 nF ceramic × 2, 10 µF electrolytic, 100 µF electrolytic, 0.1 µF ceramic × 8	—
Push-buttons	2	tactile	for TRIGGER, RESET
Breadboards	2	MB-102 830-point	—
Power supply	1	Dual ±5 V + single +5 V	Lab bench

Table B.1 Master component bill of materials.

Appendix C Additional Notes on Operating Procedure

C.1 Quick-Start Procedure

15. Apply power: confirm ±5 V to the analog board and +5 V to the digital board.
16. Place the three Ag/AgCl electrodes: positive on the left arm, negative on the right arm, reference on the right leg.
17. Verify the AD620 output on the CRO: an ECG-like waveform should be visible with V_pp in the 100–300 mV range.
18. Verify the LPF output: a cleaner waveform with the QRS R-peaks clearly above 1 V.
19. Press the RESET push-button on the digital board: the display should read 00.
20. Press the TRIGGER push-button: the 555 starts its 60 s window and the display begins counting.
21. After 60 s the display freezes at the BPM value. For a healthy resting adult this should lie in the 60–100 BPM range.

C.2 Troubleshooting

- If the comparator output stays HIGH: the reference V_ref may be too low; check the R4/R5 divider.
- If the counter increments wildly: check the comparator hysteresis resistor (1 MΩ from output to non-inverting input); without hysteresis the comparator can multi-count near threshold.
- If the display flickers but does not change: check that DEI on both CD4026s is tied to VCC.
- If the BPM reading is roughly half of the expected value: the comparator may be triggering only on the larger of two alternate R-peaks; reduce V_ref or increase gain.

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